CE 310 — Week 7 Pre-Lab Quiz

Week: 7 | Topic: Correlation, Covariance, and Scatter Plots Points: 10 questions × 1 pt each = 10 pts Submission: Complete in D2L before class begins. No partial credit for MC.

Instructions

Select the single best answer for each question. The quiz covers the Pearson correlation coefficient, covariance, scatter plots, and the Excel functions you will use in today's lab. Review the Week 7 Background reading before completing this quiz.

Questions

Q1. The Pearson correlation coefficient r between two variables always falls in which range?

(A) 0 to 1
(B) −100 to +100
(C) −1 to +1
(D) 0 to ∞

Q2. If r = −0.85 between two variables X and Y, what does this mean?

(A) As X increases, Y tends to decrease — a strong negative linear relationship
(B) As X increases, Y also increases — a strong positive linear relationship
(C) X causes Y to decrease
(D) The variables have no relationship with each other

Q3. If r = 0 between outdoor temperature and cooling energy for a particular dataset, the best interpretation is:

(A) Outdoor temperature definitely does not affect cooling energy in this building
(B) There is no linear relationship between the two variables — a non-linear relationship could still exist
(C) The dataset has fewer than 100 rows
(D) Both variables have a mean of zero

Q4. The DOE Reference Medium Office Building dataset used in CE 310 has how many rows of hourly data?

(A) 365
(B) 2,600
(C) 5,058
(D) 8,760

Q5. In the CE 310 dataset, which pair of variables has the highest Pearson r in absolute value when computed over all 8,760 hours?

(A) Zone_Temp_F and Month ($r \approx 0.14$)
(B) Zone_Temp_F and Cooling_kWh ($r \approx 0.28$)
(C) Outdoor_Temp_F and Cooling_kWh ($r \approx 0.78$)
(D) Zone_Temp_F and Outdoor_Temp_F ($r \approx 0.75$)

Q6. Which statement correctly distinguishes covariance from correlation?

(A) Covariance is always larger than correlation
(B) Covariance has units (e.g., °F × kWh); correlation is dimensionless and always falls between −1 and +1
(C) Correlation measures how two variables move together; covariance does not
(D) Covariance and correlation always give the same numerical value

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Q7. The coefficient of determination r^2 tells you:

(A) The slope of the regression line between two variables
(B) The fraction of the variance in one variable that is linearly explained by the other variable
(C) The probability that two variables are related
(D) The square root of the covariance between two variables

Q8. In a machine learning project, you compute all pairwise Pearson r values between your input features. You find that Feature A and Feature B have $r = 0.97$. The main concern is:

(A) Both features will improve the regression model equally
(B) The features are negatively correlated, which makes regression unstable
(C) Both features carry nearly identical information — including both creates multicollinearity, inflating coefficient standard errors and making the model unstable
(D) You should remove both features because $r > 0.9$ means they are noise

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Q9. In Excel, which formula correctly computes the Pearson correlation between Outdoor_Temp_F (column H, rows 2–8761) and Cooling_kWh (column I, rows 2–8761)?

(A) =PEARSON(H2:H8761, I2:I8761)
(B) =CORREL(H2:H8761, I2:I8761)
(C) =COVARIANCE.S(H2:H8761, I2:I8761)
(D) Both (A) and (B) return the same value

Q10. In Excel, =COVARIANCE.S(H2:H8761, I2:I8761) uses which denominator?

(A) n (population covariance)
(B) n − 1 (sample covariance) — the same convention as Excel's STDEV function
(C) n − 2 (used in regression residual variance)
(D) The square root of n
